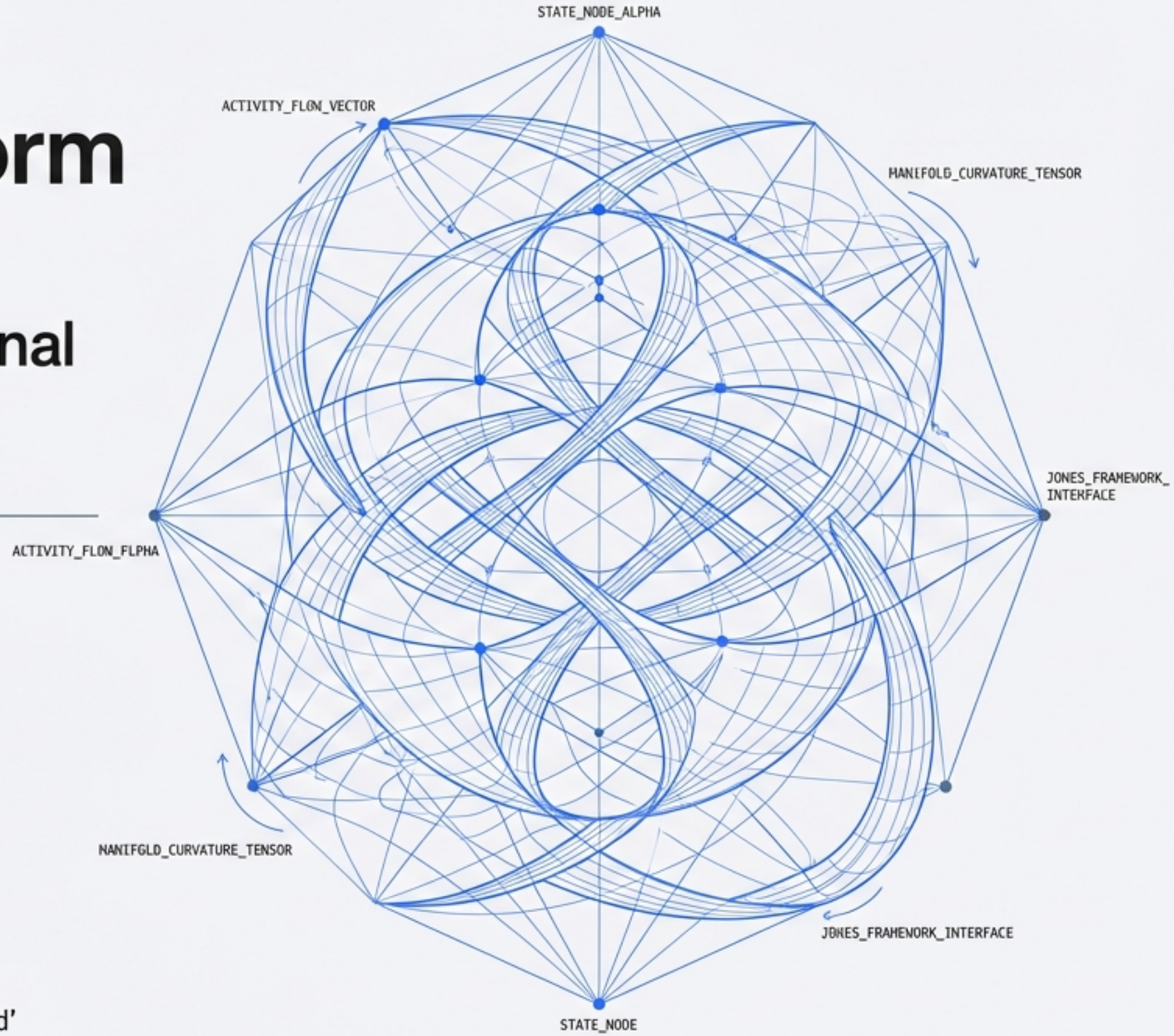
SupaTrupa: Unified Activity:State Platform

A Cognitive Command Center for State-Adaptive Computational Intelligence.

A production-ready synthesis of two engineering systems: The theoretical rigor of the Jones Framework combined with the visual power of real-time 3D dashboards.



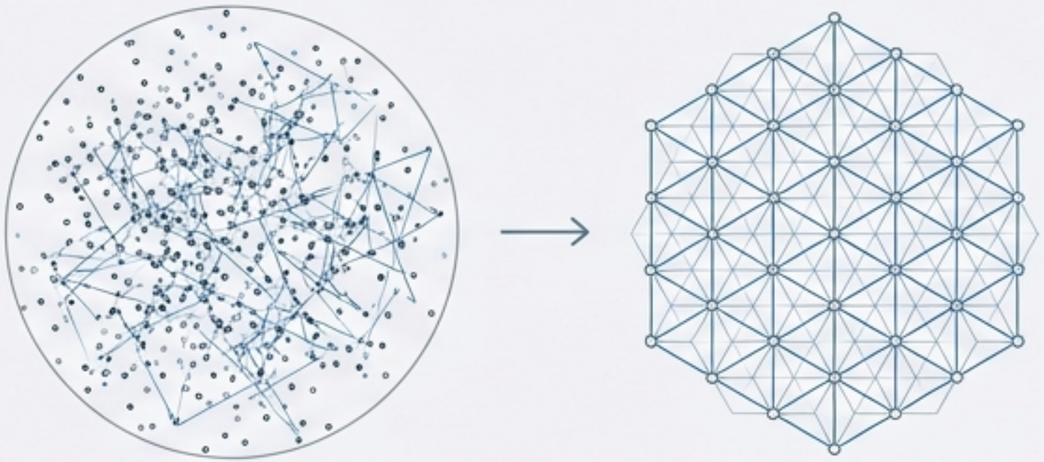
Source: Synthesized from ‘Bestest-main’ and ‘activity-state-dashboard’

Complex systems exist in distinct thermodynamic phases

The Principle

The Geometry of Being (GoB)

The system operates on the principle that the rules of physics change between regimes. We do not monitor linear trends; we monitor phase shifts.



Chaotic Regime

Structured Regime

Phase Transition: Topological Shift in System Metric

The Primitives



`ConditionState`

Atomic, immutable observations. Comparable to spin network nodes in Loop Quantum Gravity.



`ActivityState`

Macroscopic regimes defined by distinct metric tensors.

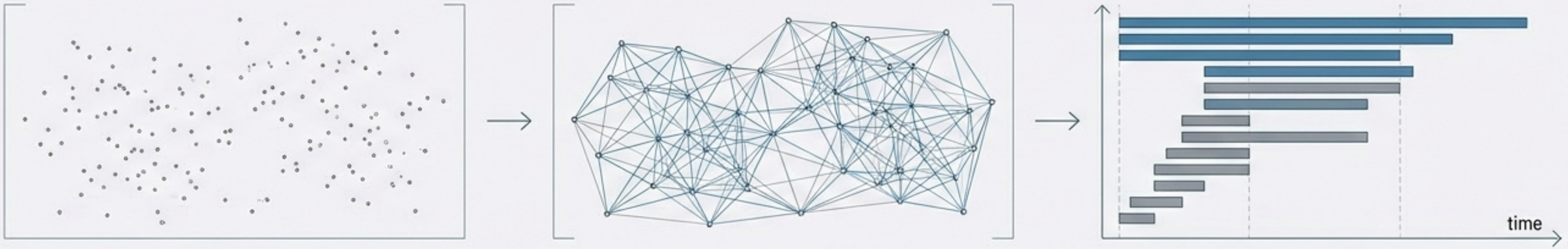


`Geodesics`

The optimal paths within the system, which curve according to the geometry of the current regime.

The system perceives structure through Persistent Homology

TDA Pipeline via Ripser



Point Cloud → Vietoris-Rips Complex → Persistence Barcode

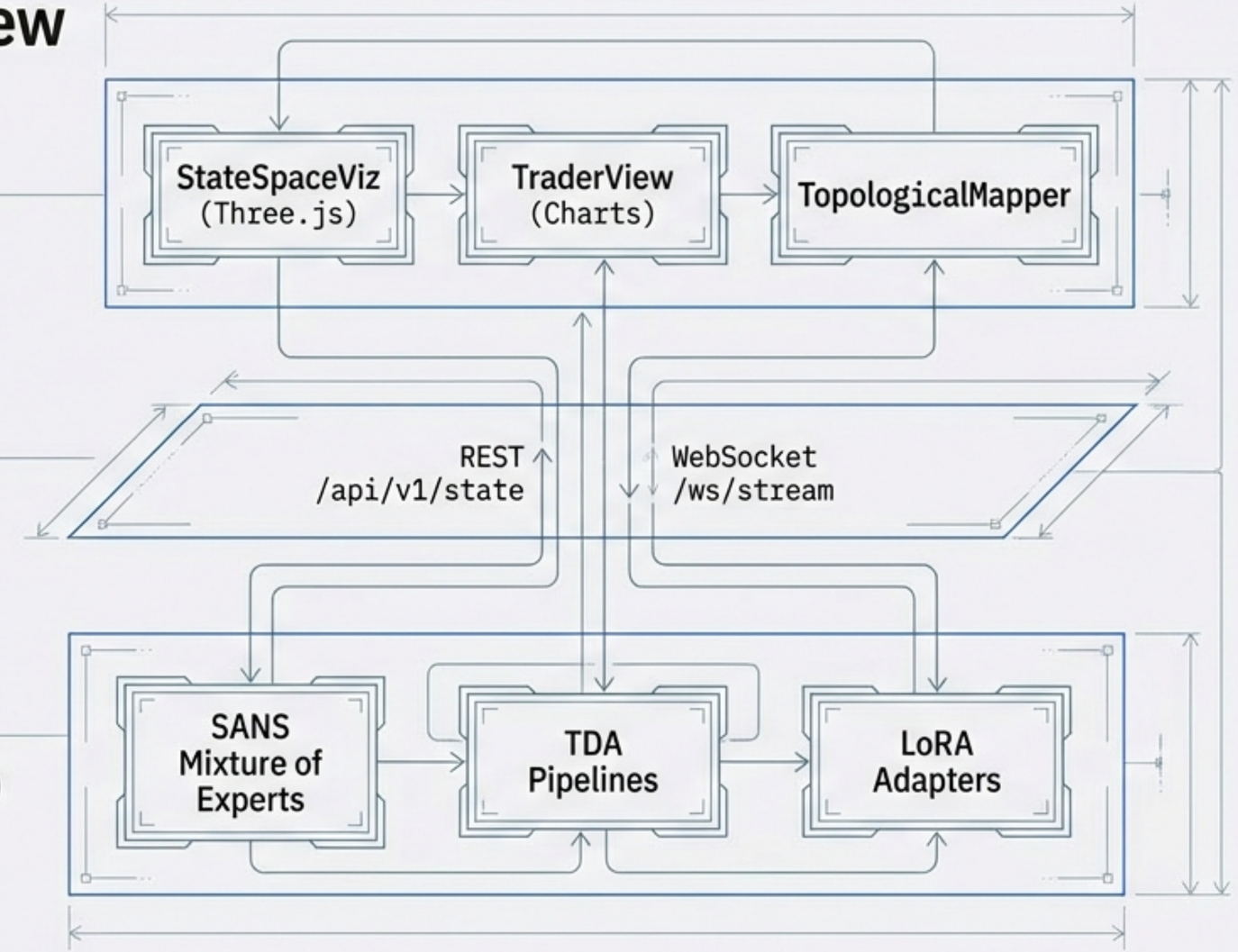
The Betti Numbers

Betti Number	Interpretation	Application
β_0	Connected Components	Fragmented markets / Disconnected states
β_1	Loops / Cycles	Oscillating patterns (Vibration analysis)
β_2	Voids	Structural gaps in the dataset

We do not just measure values; we measure the topology of the signal.

A synthesis of Backend rigor and Frontend visualization

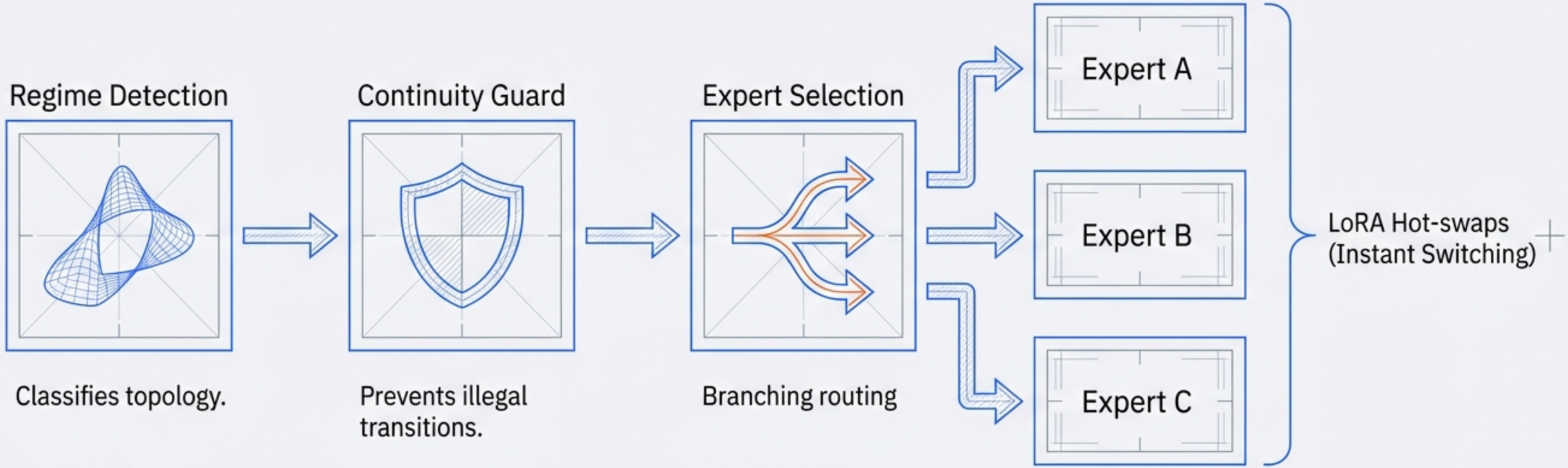
Exploded View



Integration:
Shared Type System with JSON Schemas validated connaly validated across Python and Python and TypeScript ensures **Mission Persistence**.

The Jones Framework drives the Computational Engine

SANS Architecture: Symbolic Abstract Neural Search



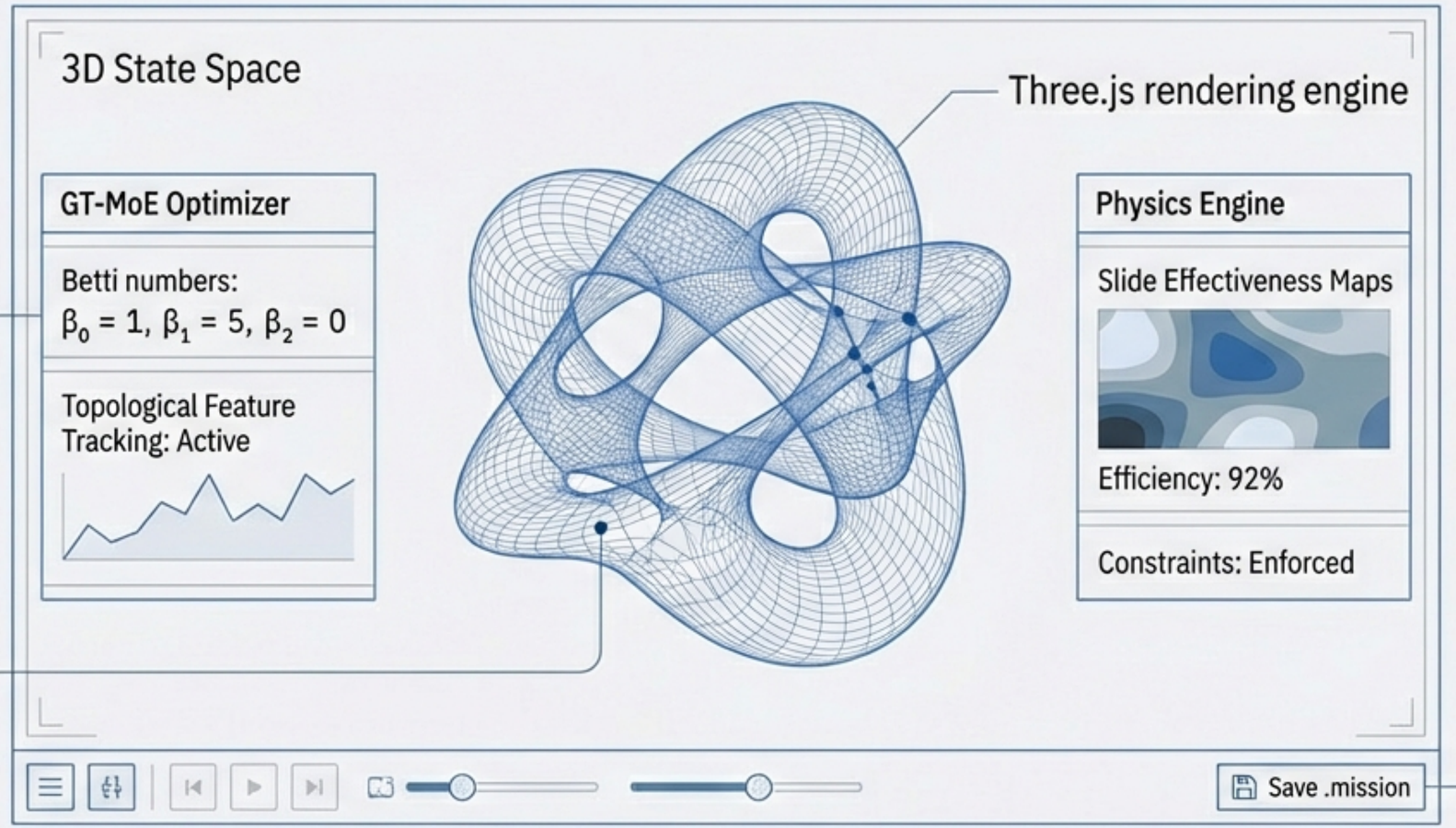
Python-based backend designed for State-Adaptive Computational Intelligence.
Utilizing `Mixture of Experts` to route data to regime-specific models.

```
> jones_framework.route(data)
-> Expert(B).process_with_lora()
>
```

Visualization of high-dimensional manifolds in real-time

React 19 + Radix UI
IBM Plex Sans
IBM Plex Mono

React 19 + Radix UI
IBM Plex Sans
IBM Plex Mono



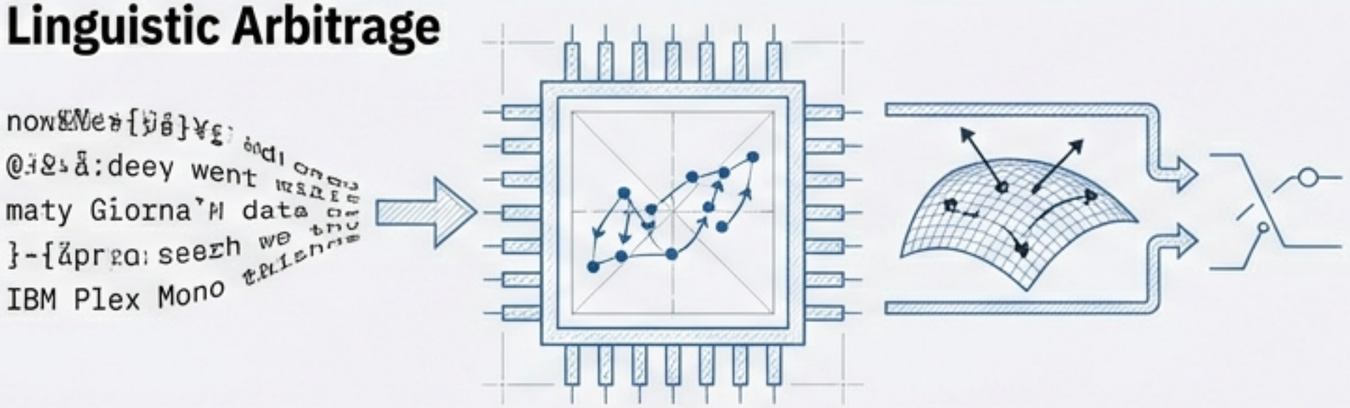
Stack:

React 19,
Radix UI,
Tailwind CSS,
Framer Motion.
IBM Plex Mono

Mission Persistence
IBM Plex Sans
IBM Plex Mono

Linguistic Arbitrage and Hardware Acceleration

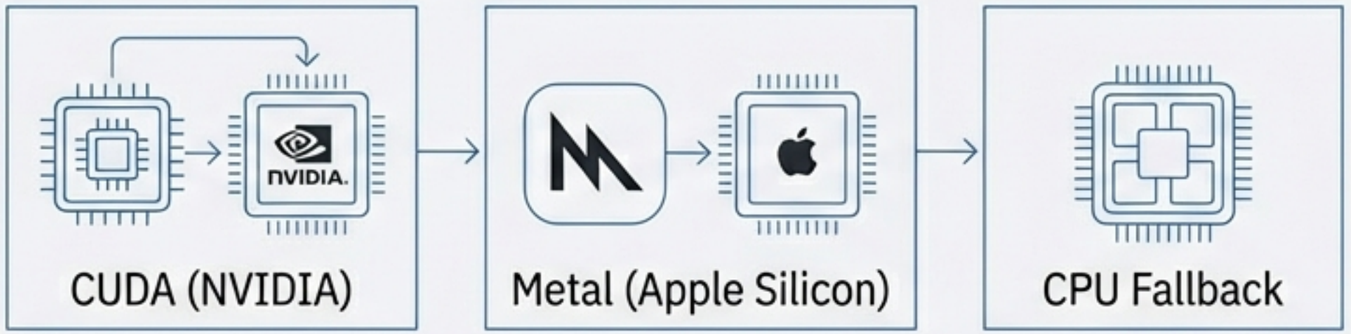
Linguistic Arbitrage



Sentiment Vectors drive regime shifts based on non-numerical data.

NLP-based **kinetic triggers** bridge the **Manifold (Math)** with the **Sentiment (Language)**.

Hardware Agnostic Acceleration




```
JONES_DEVICE = auto
```

Automatic detection of available compute resources ensures deployment flexibility.

Les Logic: Applying the Geometry of Being to BHA Optimization

Bottom Hole Assembly Optimization Methodology

Traditional / Reactive

Threshold Check

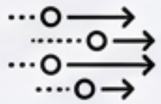


ALERT
(Damage Started)

Cost: >$500k

$↑

Sensor Data

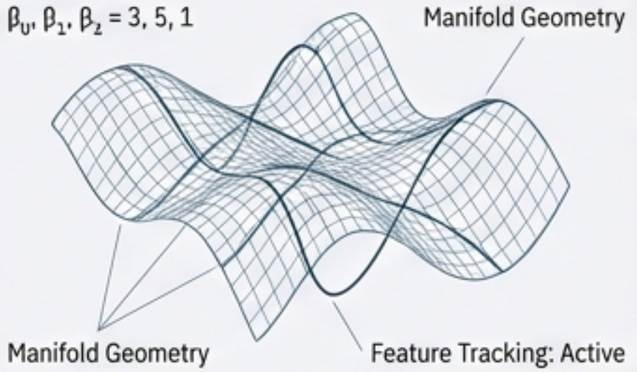


IBM Plex Mono

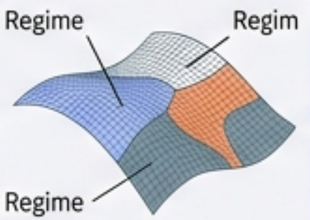
Problem: Traditional methods are reactive and threshold-based.

Les Logic / Predictive

Topological Analysis



Regime Detection



Recommendation (Prevention)

Cost: $0

0↓

Solution: Les Logic prevents damage by detecting dysfunction topology before thresholds are breached.

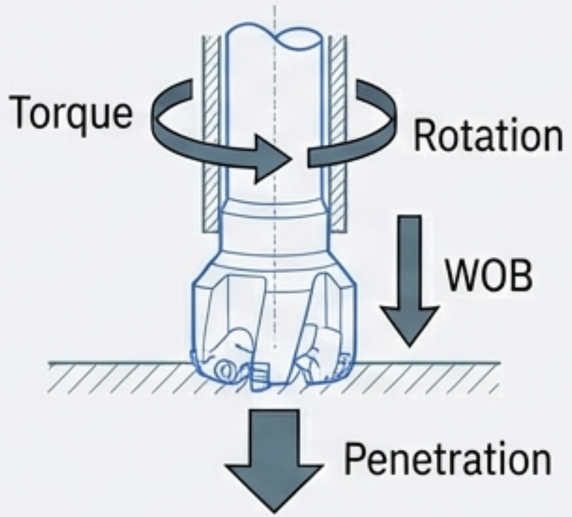
The Three Axioms of Drilling Intelligence

01 Axiomatic Alignment

Every measurement must connect to physical behavior.

Example:

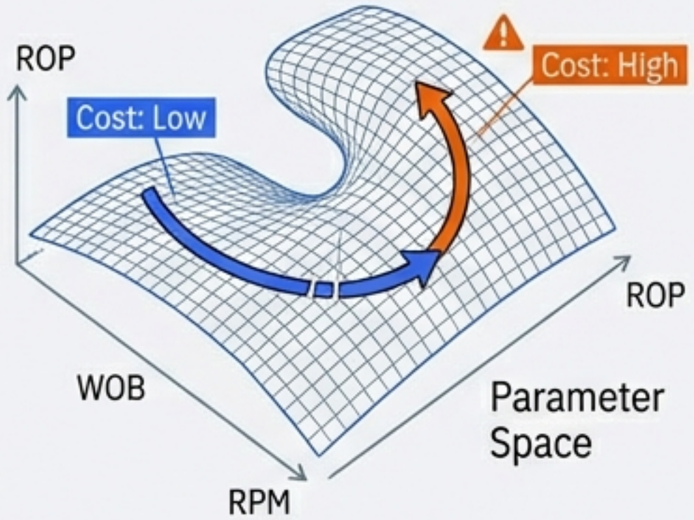
- Torque causes Rotation;
- WOB causes Penetration.



02 Non-Euclidean Topology

The drilling state space is curved.

Changes in ROP “cost” different amounts depending on the location in the parameter space.



03 Cognitive Ergonomics

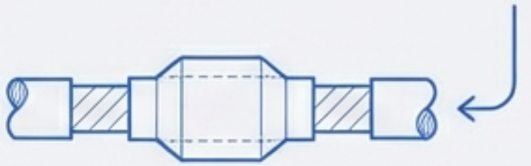
Outputs must be actionable for the driller.

Bad Output:

```
Bad Output:
"β₁ increased"
```

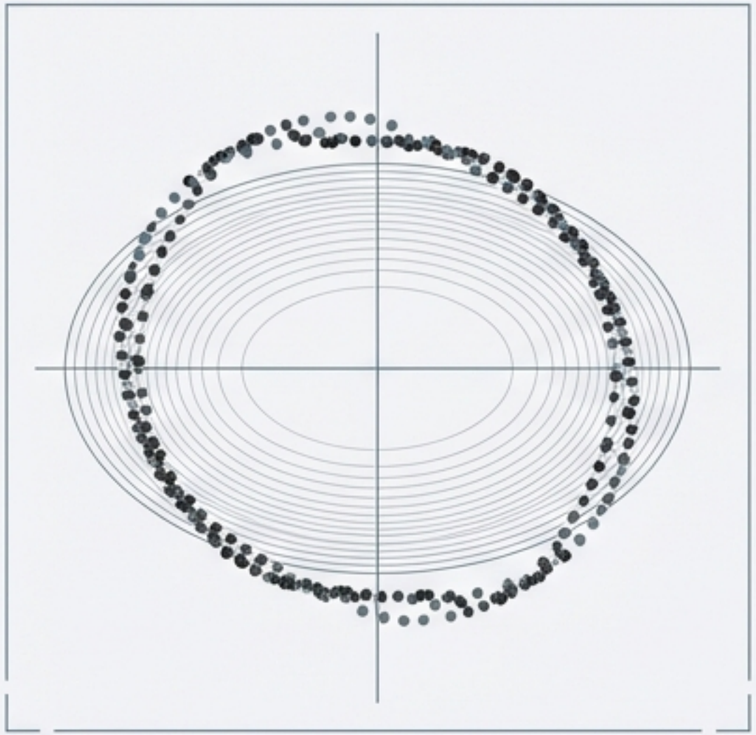
Good Output:

```
"Add 1 stabilizer"
```



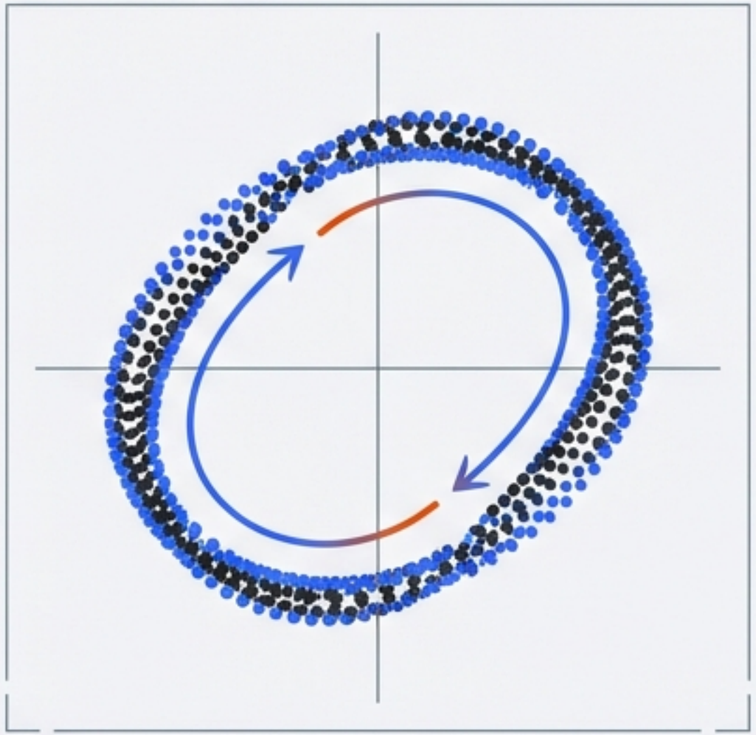
Regime detection via point clouds and oscillation

Whirl



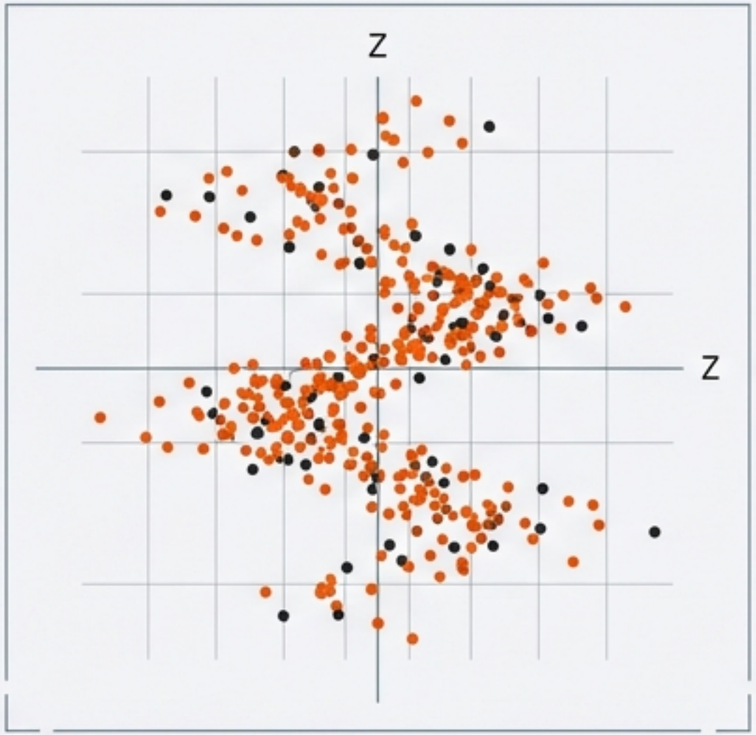
Identified via geometric eccentricity.

Stick-Slip



Identified via torsional resonance loops (β_1 cycles).

Bit Bounce



Identified via axial instability.

Riemannian Metric quantifies the ‘cost’ of operations. Ricci curvature indicates changes in formation difficulty.

Physics-based recommendations override reactive alerts

Detected Regime	Root Cause	Action (Remediation)
Whirl	BHA eccentricity	Add stabilizer ↗
Stick-Slip	Torsional resonance	Reduce motor bend angle ↗
Bit Bounce	Axial instability	Increase flow restrictor ↗

Dysfunctions are topological shapes in phase space. We fix ⁞'. We fix the shape, we fix the drilling.

A production-ready tech stack

Prerequisites & Languages

- Python 3.9+
- Node.js 18+
- pnpm
- Python (79.0%)
- TypeScript (19.7%)

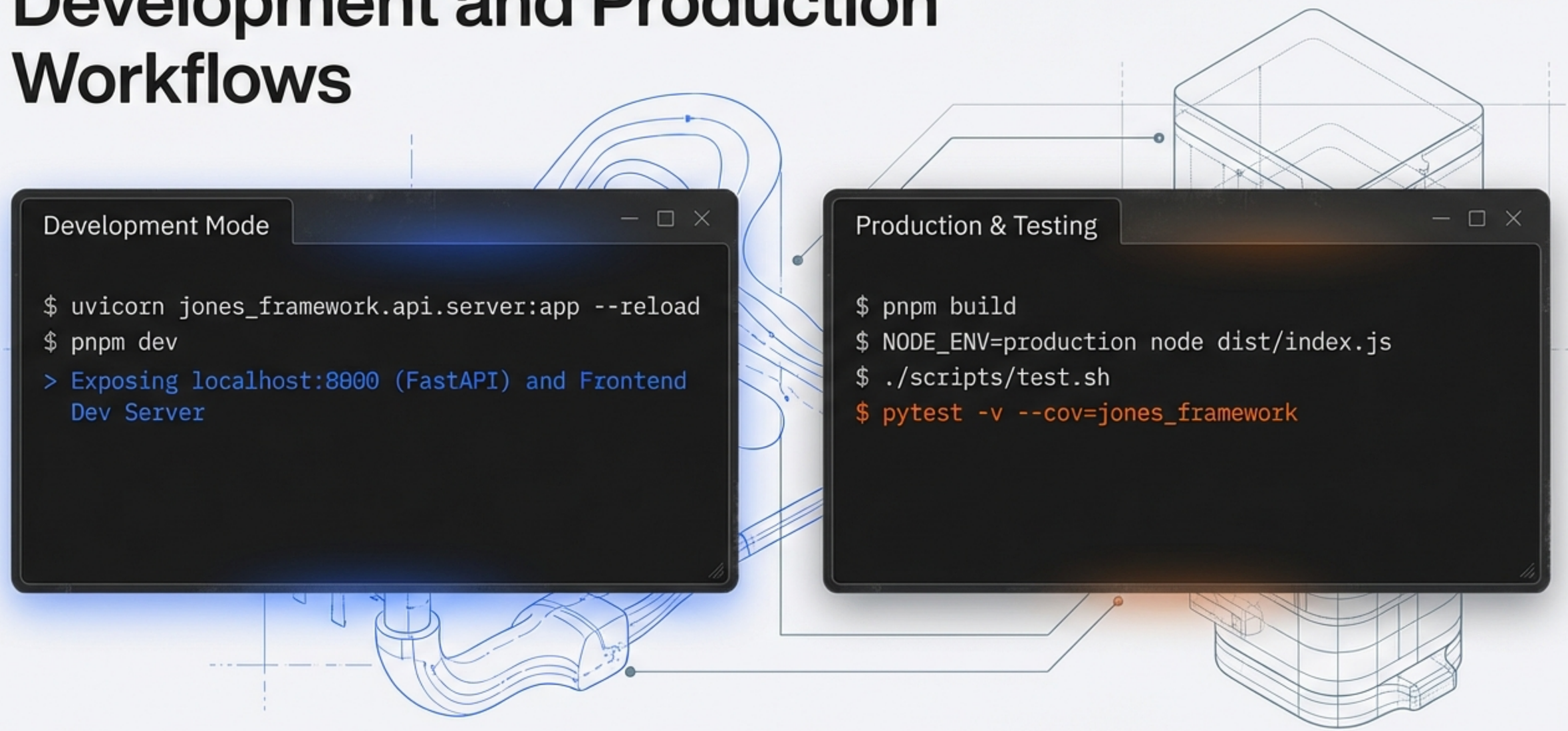
Installation

```
git clone <repo>

# Backend
pip install -e '.[all]'

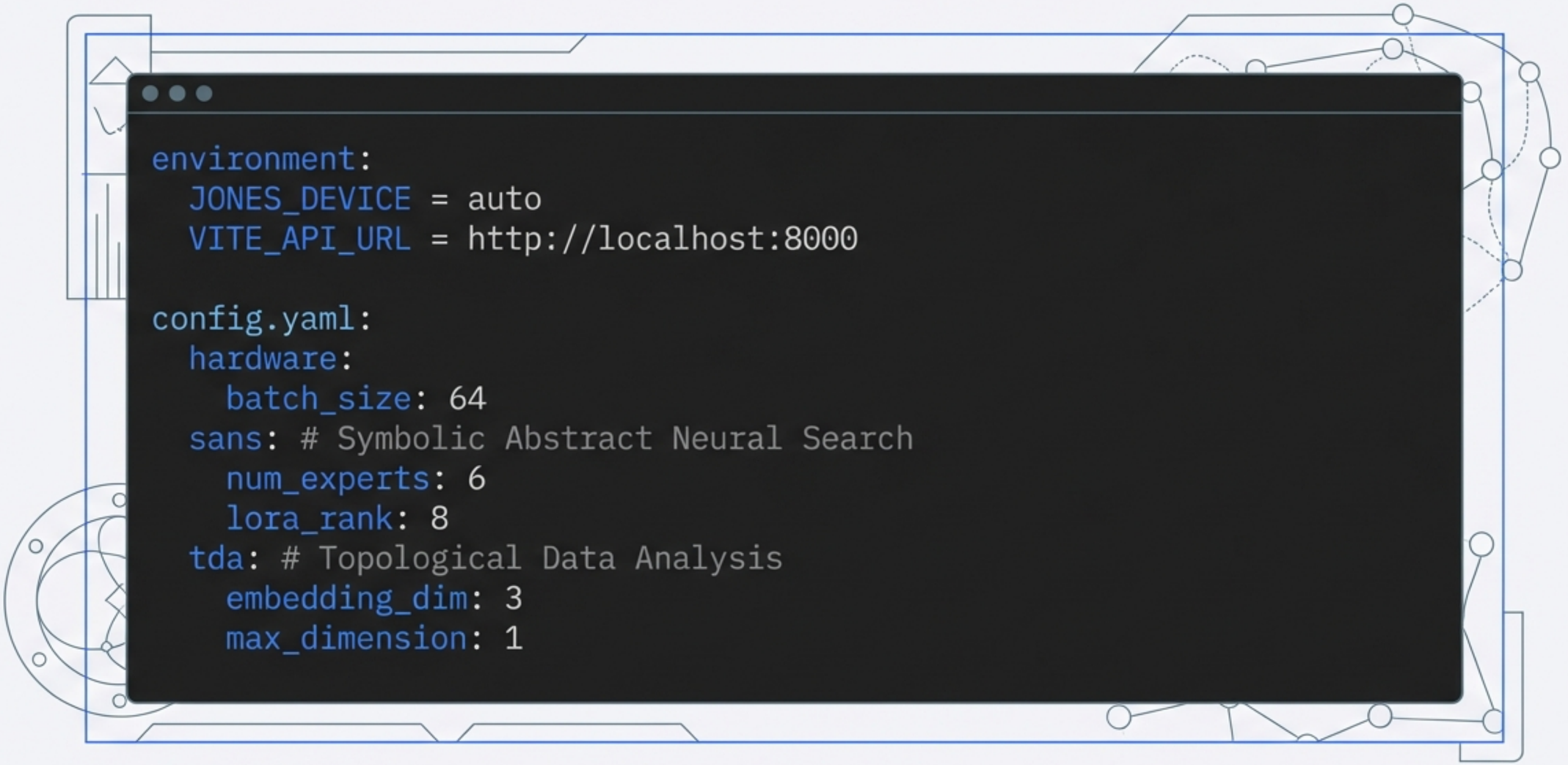
# Frontend
pnpm install
```

Development and Production Workflows



Configuration via Environment or YAML

Fully customizable embedding dimensions and expert counts.

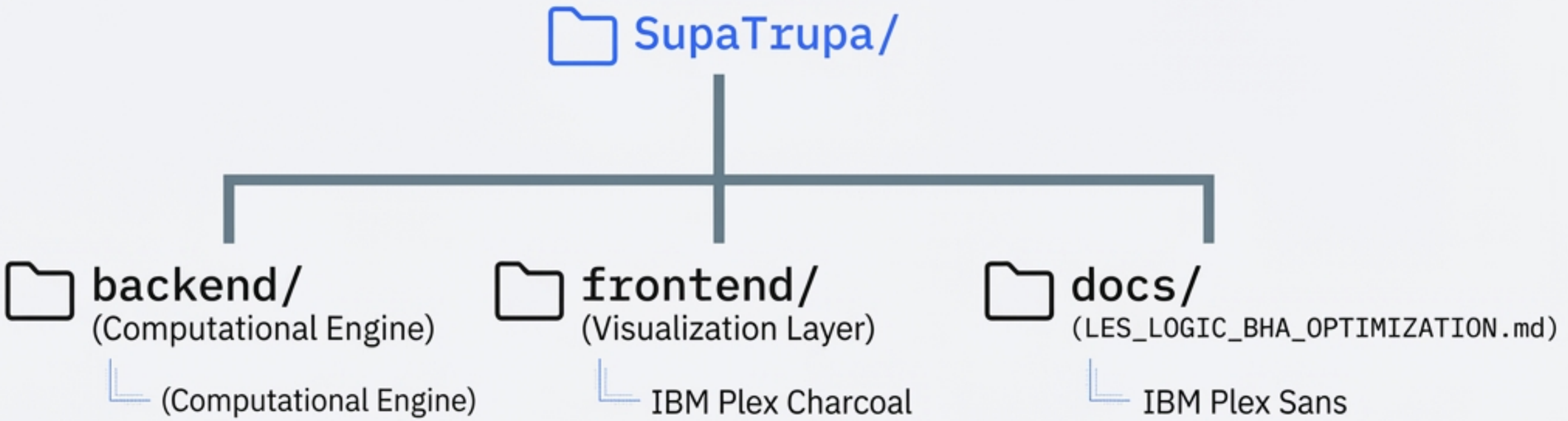


```
environment:
  JONES_DEVICE = auto
  VITE_API_URL = http://localhost:8000

config.yaml:
  hardware:
    batch_size: 64
  sans: # Symbolic Abstract Neural Search
    num_experts: 6
    lora_rank: 8
  tda: # Topological Data Analysis
    embedding_dim: 3
    max_dimension: 1
```

SupaTrupa: The Future of Industrial AI





Synthesized from Jones Framework and Activity-State Dashboard.
License: MIT License
Ready for deployment. Clone the repository to begin.